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Assignment #3:

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2.7

a) $q'_x = \frac{q_x}{A} = -k \frac{dT}{dx}$
 $k = 120 \frac{W}{m \cdot K}$ $A = 1.2m \times 10^{-2} m^2$
 $T_1 = 500 K$ $T_2 = 700 K$
 $\frac{dT}{dx} = \frac{500 - 700}{1.2 \times 10^{-2}} = -16666.67 \frac{K}{m}$
 $q'_x = -120 \frac{W}{m \cdot K} \times -16666.67 \frac{K}{m} = 2 \times 10^6 \frac{W}{m^2}$

b) $\frac{dT}{dx} = \frac{500 - 700}{1.2 \times 10^{-2}} = -16666.67 \frac{K}{m}$
 $q'_x = -120 \frac{W}{m \cdot K} \times -16666.67 \frac{K}{m} = 2 \times 10^6 \frac{W}{m^2}$

c) $\frac{dT}{dx} = \frac{700 - 500}{1.2 \times 10^{-2}} = 16666.67 \frac{K}{m}$
 $q'_x = 120 \frac{W}{m \cdot K} \times 16666.67 \frac{K}{m} = 2 \times 10^6 \frac{W}{m^2}$

2.19

$$T(x) = \frac{qL^2}{2k} \left(1 - \frac{x^2}{L^2}\right) + \frac{T_{s1} - T_{s2}}{2} \frac{x}{L} + \frac{T_{s1} + T_{s2}}{2}$$

$$\frac{\partial}{\partial x} \left(k \frac{dT}{dx} \right) + \frac{\partial}{\partial y} \left(k \frac{dT}{dy} \right) + \frac{\partial}{\partial z} \left(k \frac{dT}{dz} \right) + q = 0$$

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + q = 0$$

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) = -q$$

$$k \frac{dT}{dx} = -q x$$

$$\int \frac{dT}{dx} = \int \frac{-q}{k} dx$$

$$\int \frac{dT}{dx} = \int \frac{-q}{k} dx + C_1$$

$$T(x) = -\frac{qx^2}{2k} + C_1 x + C_2$$

$$T(x) = -\frac{qx^2}{2k} + \left(\frac{T_{s1} - T_{s2}}{-2L} \right) x + \frac{T_{s1} - T_{s2}}{2} + \frac{T_{s1} + T_{s2}}{2}$$

$$= \frac{qL^2}{2k} \left(1 - \frac{x^2}{L^2} \right) - \frac{T_{s1} - T_{s2}}{2} \frac{x}{L} + \frac{T_{s1} + T_{s2}}{2}$$

$$T(x) = \frac{qL^2}{2k} \left(1 - \frac{x^2}{L^2} \right) + \frac{T_{s2} - T_{s1}}{2} \frac{x}{L} + \frac{T_{s1} + T_{s2}}{2}$$

$$q'_x = -k \frac{dT}{dx} = -k \left[-\frac{q}{k} x + C_1 \right]$$

$$q'' = q_x \left[\frac{T_1 - T_2}{L} \right] k$$

BC #1 $T = T_{s1}$ @ $x = 0$

BC #2 $T = T_{s2}$ @ $x = L$

① $-T_{s1} = -\frac{qL^2}{2k} - C_1 L + C_2$

② $-T_{s2} = -\frac{qL^2}{2k} + C_1 L + C_2$

① - ②

$$T_{s1} - T_{s2} = -2C_1 L$$

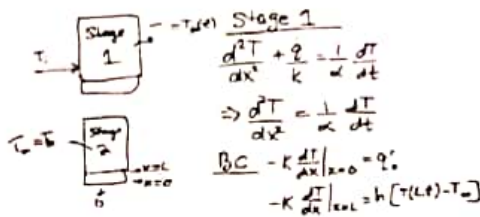
$$C_1 = \frac{T_{s1} - T_{s2}}{-2L}$$

$$T_{s2} = -\frac{qL^2}{2k} + \frac{T_{s1} - T_{s2}}{-2} + C_2$$

$$T_{s2} - \frac{qL^2}{2k} + \frac{T_{s1} - T_{s2}}{2} = C_2$$

$$C_2 = \frac{qL^2}{2k} + \frac{T_{s1} + T_{s2}}{2}$$

2.20



Stage 2

$$\frac{d^2T}{dx^2} = 0$$

$$BC: -k \frac{dT}{dx} \big|_{x=0} = q'_x$$

$$T(0) = T_\infty$$

2.40:

$$T(r) = \frac{C_1}{r} + C_2$$

from $r_1 \rightarrow r_2$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{\partial T}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(k \sin \theta \frac{\partial T}{\partial \theta} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \phi} \left(k \sin \theta \frac{\partial T}{\partial \phi} \right) + q = \rho c_p \frac{\partial T}{\partial t}$$

$$= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) = \frac{1}{\alpha} \frac{\partial T}{\partial t}$$

$$T(r) = \frac{C_1}{r} + C_2$$

$$\frac{1}{\alpha} \frac{\partial T}{\partial t} = -\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{C_1}{r^2} \right) = 0$$

Steady state exists

$$q'_r = -k \frac{\partial T}{\partial r} = k \frac{C_1}{r^2}$$

$$Q = 4\pi r^2 q'_r \rightarrow 4\pi k C_1$$

Q independent of r